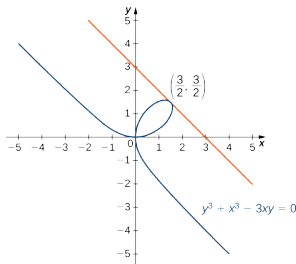


Week 05, Lecture 2 Implicit Differentiation; Exponential Functions

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Today, in Engineering Calculus...

- 1 Implicit Functions
- 2 Implicit Differentiation
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 - The derivative of e^x
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 - Properties
 - The natural logarithm
 - Derivative of $\ln(x)$

8 Exercises

Slides are on canvas, and
at [niallmadden.ie/2627-
MA140](http://niallmadden.ie/2627-MA140)

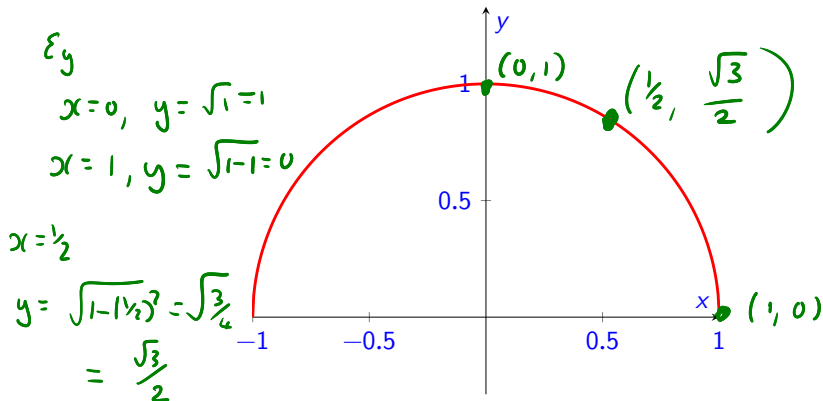


See also: Sections 3.8 (Implicit Differentiation) and 3.9 (Derivatives of Exponential and Logarithmic Functions) of **Calculus** by Strang & Herman: <https://openstax.org/books/calculus-volume-1/pages/3-8-implicit-differentiation> and <https://openstax.org/books/calculus-volume-1/pages/>

Implicit Functions

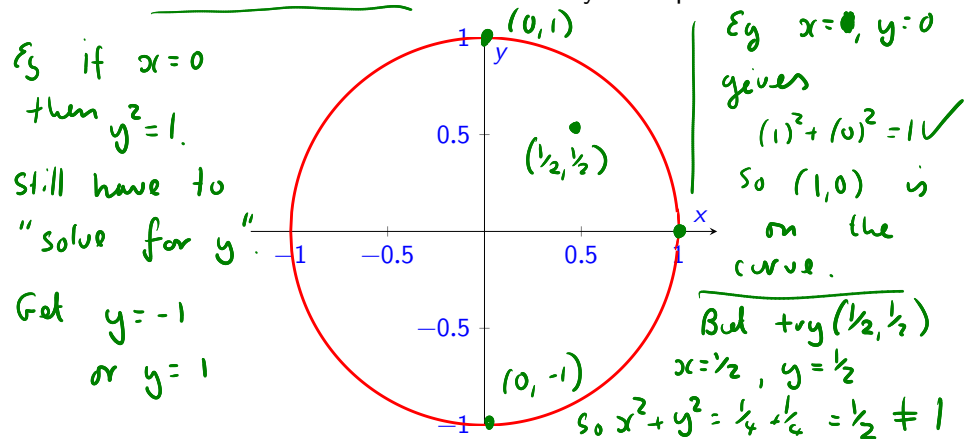
To date, most functions we have studied have been **explicitly** defined. Such functions can be written as $y = f(x)$: given a value of x we can substitute it into $f(x)$ to get the corresponding value of y

Example: $y = \sqrt{1-x^2}$. Domain is $x \in [-1, 1]$



Implicit Functions

However, sometimes we are given an equation involving x and y where these two terms are not “separated” entirely; e.g., $x^2 + y^2 = 1$. Here y is **implicitly** defined: for any pair (x, y) we can check if it is on the curve described by the equation.



Implicit Differentiation

Implicit equations define curves; we can use a technique called **implicit differentiation** to find tangents to these curves.

$$\frac{d}{dx}(x^2) = 2x$$

Method:

1. Differentiate both sides of the equation, with respect to x , keeping in mind that y is a function of x , using the Chain Rule where needed.

2. Solve for dy/dx .

Suppose we want to differentiate y^2 ,
where y is some function of x .
Use Chain Rule : $\frac{df}{dx} = \frac{df}{du} \cdot \frac{du}{dx}$ where $f(x) = u(v(x))$

$$\frac{d}{dx}[y^2] = \underbrace{\frac{d}{dy}[y^2]}_{2y} \cdot \underbrace{\frac{dy}{dx}}_{\frac{dy}{dx}} = 2y \frac{dy}{dx}$$

Implicit Differentiation [wrt = "with respect to"]

Example

If y is defined by $x^2 + y^2 = 1$, find $\frac{dy}{dx}$.

The curve is defined as $x^2 + y^2 = 1$.
Differentiate both sides w.r. to x

$$\frac{d}{dx} [x^2 + y^2] = \frac{d}{dx} [1]$$

$$\Rightarrow \frac{d}{dx} [x^2] + \frac{d}{dx} [y^2] = 0$$

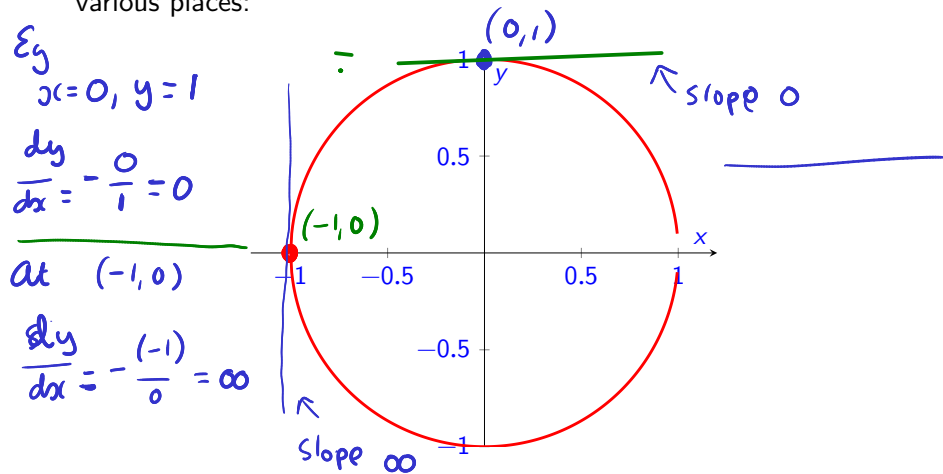
$$\Rightarrow 2x + 2y \frac{dy}{dx} = 0 \quad (\text{From slide 5})$$

$$\Rightarrow 2y \frac{dy}{dx} = -2x \Rightarrow$$

$$\frac{dy}{dx} = -\frac{2x}{2y} = -x/y$$

Implicit Differentiation

Now we know that if $x^2 + y^2 = 1$, then $\frac{dy}{dx} = -\frac{x}{y}$. We can check that this relates to the slope of the tangents to this curve at various places:



Implicit Differentiation

Now we know that if $x^2 + y^2 = 1$, then $\frac{dy}{dx} = -\frac{x}{y}$. We can check that this relates to the slope of the tangents to this curve at various places:

Try $(x, y) = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$

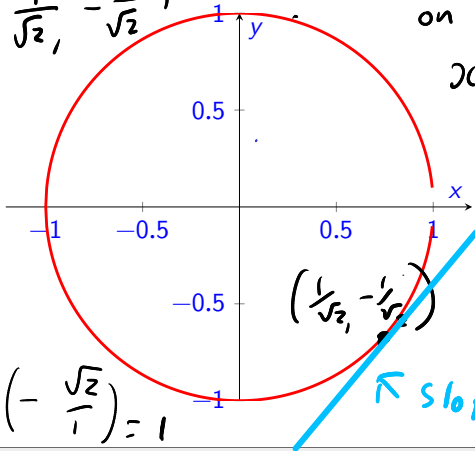
Slope of tangent at (x, y) is

$$\frac{dy}{dx} = -\frac{x}{y}$$

$$= -\left(\frac{1}{\sqrt{2}}\right)\left(-\frac{\sqrt{2}}{1}\right) = 1$$

check this is on the curve:

$$\begin{aligned}x^2 + y^2 &= \\ \left(\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2 &= \\ &= \frac{1}{2} + \frac{1}{2} = 1 \quad \checkmark\end{aligned}$$



Implicit Differentiation

Taken from Textbook.



Find the tangent to the curve $x^2 + y^2 = 25$, at the point $(3, -4)$.

First check that the point $(x, y) = (3, -4)$ is on the curve:

$$(3)^2 + (-4)^2 = 9 + 16 = 25, \text{ as required.}$$

Now differentiate:

$$\frac{d}{dx}[x^2 + y^2] = \frac{d}{dx}[25] \implies \frac{d}{dx}[x^2] + \frac{d}{dx}[y^2] = 0$$

$$\implies \underline{2x} + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}.$$

So the slope of the tangent to the curve at $x_1 = 3$ and $y_1 = -4$ is

$$M = -x/y = 3/4. = (-3/(-4)) = 3/4.$$

The equation of the tangent comes from $y - y_1 = M(x - x_1)$. For our values of $x_1 = 3$, $y_1 = -4$ and $M = 3/4$, we get

$$y = (-4) + (3/4)(x - 3). \text{ That is } y = \frac{3}{4}x - \frac{25}{4}.$$

Implicit Differentiation

Find the tangent to the curve $y^3 + x^3 - 3xy = 0$, at the point $(3/2, 3/2)$.

First check: is $(x, y) = (\frac{3}{2}, \frac{3}{2})$ on the curve?

$$\begin{aligned} y^3 + x^3 - 3(x)(y) &= \left(\frac{3}{2}\right)^3 + \left(\frac{3}{2}\right)^3 - 3\left(\frac{3}{2}\right)\left(\frac{3}{2}\right) \\ &= \underbrace{\frac{27}{8} + \frac{27}{8}}_{\frac{27}{4}} - \frac{27}{4} = 0 \quad \checkmark \end{aligned}$$

Implicit Differentiation

Find the tangent to the curve $y^3 + x^3 - 3xy = 0$, at the point $(3/2, 3/2)$.

Next: find $\frac{dy}{dx}$.

$$\frac{d}{dx} [y^3] + \frac{d}{dx} [x^3] - \frac{d}{dx} [3xy] = 0$$

$$\frac{d}{dx} [y^3] = 3y^2 \cdot \frac{dy}{dx} \quad (\text{Chain Rule})$$

$$\frac{d}{dx} [x^3] = 3x^2$$

$$\frac{d}{dx} [xy] = u \cdot v \quad \begin{matrix} u = x & v = y \\ u' = 1 & v' = \frac{dy}{dx} \end{matrix} \Rightarrow \frac{d}{dx} [xy] = x \frac{dy}{dx} + y$$

Implicit Differentiation

Find the tangent to the curve $y^3 + x^3 - 3xy = 0$, at the point $(3/2, 3/2)$.

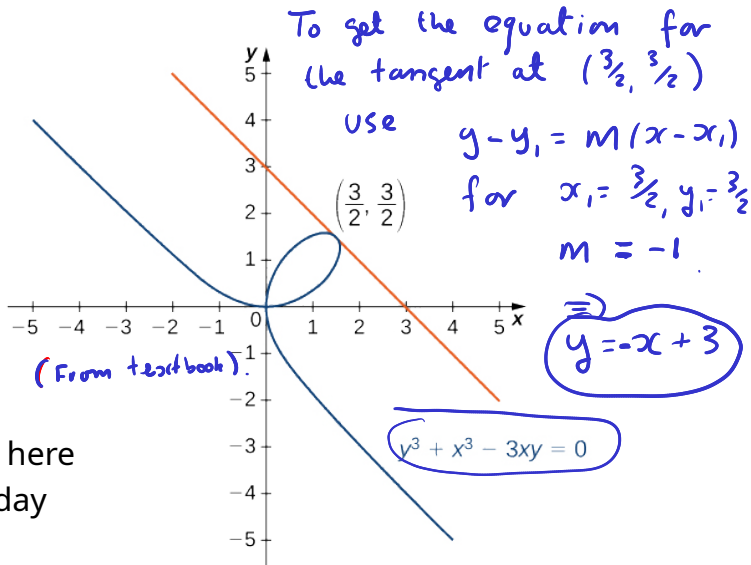
$$\text{Then } \frac{d}{dx} [y^3 + x^3 - 3xy] = 0$$

$$\Rightarrow 3y^2 \frac{dy}{dx} + 3x^2 - 3\left(x \frac{dy}{dx} + y\right) = 0.$$

$$\text{Find } \frac{dy}{dx} : \quad \frac{dy}{dx} (3y^2 - 3x) = 3y - 3x^2$$

$$\text{So } \frac{dy}{dx} = \frac{y - x^2}{y^2 - x} \quad \left| \begin{array}{l} \text{Then if } (x, y) = (3/2, 3/2) \\ \text{we get } \frac{dy}{dx} = -1. \text{ (check!)} \end{array} \right.$$

Implicit Differentiation



Finished here
Wednesday

Exponential functions

Earlier in this course we met functions such as $y = x^2$; this is a **power** function.

Now we consider **exponential functions**, such as $y = 2^x$.

Such functions occur in many applications. For example: if I invest €100 with an annual interest rate of 20%, then after x years, I will have € $100 \times (1.2)^x$. **Why?**

$$\frac{d}{dx} (y^3) = \frac{d}{dx} ((y(x))^3)$$

$$u(v) = y^3$$

$$v(x) = y(x)$$

$$f(x) = u(v(x))$$

$$\frac{df}{dx} = \frac{du}{dv} \cdot \frac{dv}{dx} = 3y \frac{dy}{dx}$$

Exponential functions

Exponential functions grow quite fast: if my investment is indeed worth $f(x) = 100 \times (1.2)^x$ euros after x years, then...

- ▶ After 1 year, I have €120
- ▶ After 10 years, I have €619.17
- ▶ After 20 years, I have €3,833.80
- ▶ After 25 years, I have €9,539.60
- ▶ After 50 years, and 190 days, I'll be a millionaire!

Properties of exponents

Here I remind you of some properties of exponents that you should already know: for any positive numbers a and b , and for all x and y ,

$$1. \quad b^x b^y = b^{x+y}$$

$$2. \quad \frac{b^x}{b^y} = b^{x-y}$$

$$3. \quad (b^x)^y = b^{xy}$$

$$4. \quad (ab)^x = a^x b^x$$

$$5. \quad \left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$$

There is more information on this in [Section 1.5 of the textbook \(Exponential and Logarithmic Functions\)](#). ([← link to book](#))

The number e

The number e is a mathematical constant (similar, in a sense, to the way that $\pi \approx 3.14159$ is too).

The value of e is roughly 2.7182818284.

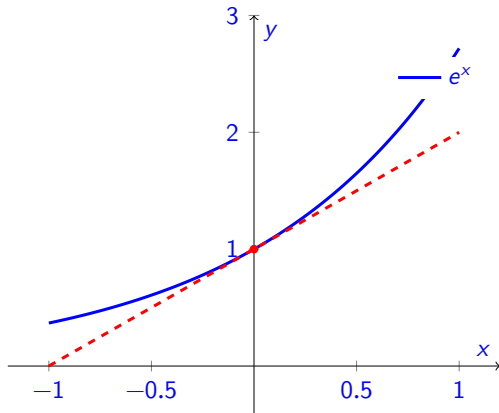
It is often called **Euler's Number** after Leonard Euler, who did not discover it: that was (probably) Jacob Bernoulli in 1683 while studying compound interest. Or maybe 100 years earlier by John Napier.

It has some very interesting and important properties...

Natural Exponential Function

The Natural Exponential Function

The Natural Exponential Function is $f(x) = e^x$. It is special for many reasons, including the its tangent at $x = 0$ has slope 1.



Let's assume that e is the number for which, if $f(x) = e^x$, then $f'(0) = 1$. Using the limit definition of the derivative, this means

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{e^{0+h} - e^0}{h} = \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1.$$

Now we use this to work out $\frac{d}{dx}[e^x]$.

So now we know that

$$\frac{d}{dx}e^x = e^x.$$

That is e^x is the function that is its own derivative!!!

Example

Compute the derivative of $f(x) = e^{\sin(x)}$

Logarithms

Suppose that $y = f(x)$ is an **exponential** function; that is: $y = b^x$ for some $b > 0$ (and excluding $x = 1$).

Its **inverse** is called a **logarithmic function**, denoted $\log_b$

$$\text{If } y = b^x \quad \text{then} \quad \log_b(y) = x.$$

Examples

- ▶ $\log_2(8) = 3$
- ▶ $\log_{10}(100) = 2$
- ▶ $\log_e(e^x) = x$

Properties of Logarithms

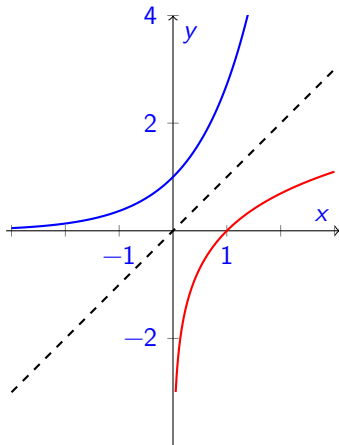
If $a, b, c > 0$ and $b \neq 1$, then

$$\blacktriangleright \log_b(ac) = \log_b(a) + \log_b(c)$$

$$\blacktriangleright \log_b\left(\frac{a}{c}\right) = \log_b(a) - \log_b(c)$$

$$\blacktriangleright \log_b(a^r) = r \log_b(a)$$

We denote $\log_e(x)$ as $\ln(x)$



$$\frac{d}{dx} \ln(x) = \frac{1}{x}$$

Why?

Example:

Find the derivative of $f(x) = \ln(x^2 + 2x + 3)$.

Exercises

Exercise 5.2.1

Find the equation of the tangent to the curve defined by $x^2 - y^2 = 16$ at the point $(5, 3)$.

For more exercises on Implicit Differentiation, see <https://openstax.org/books/calculus-volume-1/pages/3-8-implicit-differentiation>

Exercise 5.3.1 [2019 exam, Q2(b)(i)]

Differentiate $f(x) = e^{\sin(x)} \cos x$.

Exercise 5.3.2 [2023 exam, Q2(a)(i)]

Differentiate $f(x) = xe^{\sin(x)}$.